Topic: Vector Spaces

Question 1. Consider $\mathcal{V} = \mathbb{R}^* = \{\boldsymbol{r} \in \mathbb{R} : \boldsymbol{r} \geq 0\}$ with vector addition $\boldsymbol{a} \oplus \boldsymbol{b} = \sqrt{\boldsymbol{ab}}$ for all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^*$, and scalar multiplication $\lambda \otimes \boldsymbol{a} = \lambda^2 \boldsymbol{a}$ for $\boldsymbol{a} \in \mathbb{R}^*, \lambda \in \mathbb{R}$.
In this question, check which axiom holds for the above operations $\oplus$ and $\otimes$. If an axiom does not hold, give an example to illustrate it.

- Is $\mathcal{V}$ **closed closed under** $\oplus$?
- Is $\mathcal{V}$ **closed under** $\otimes$?
- Is $\oplus$ **is commutative?**
- Is $\oplus$ **is associative**?
- Is there an **additive identity**?
- Is there an **additive inverse**?
- Is $\otimes$ **is associative**?
- Is there an **multiplicative identity**?
- Does **Distributive-1** axiom hold?
- Does **Distributive-2** axiom hold?

Question 2. Consider $\mathbb{R}^2$ a vector space with the following vector addition and scalar multiplication. Show that they are NOT vector spaces. In other words, point out the axiom that fails by providing a counter example.

(a) The vector addition is the usual componentwise addition, and the scalar multiplication is $\lambda \otimes (a_1, a_2) = (a_1, 0)$.

(b) The vector addition is $(a_1, a_2) \oplus (b_1, b_2) = (a_1 + 2b_1, a_2 + 3b_2)$ and the scalar multiplication is the usual componentwise multiplication.

(c) The vector addition is the usual componentwise addition and the scalar multiplication is

$$\lambda \otimes (a_1, a_2) = \begin{cases} (0, 0), & \text{if } \lambda = 0, \\ \left(\lambda a_1, \frac{1}{\lambda} a_2\right), & \text{otherwise.} \end{cases}$$

Question 3. Let $\mathbb{R}^+ = \{\boldsymbol{x} \in \mathbb{R} : \boldsymbol{x} > 0\}$ be the set of positive real numbers. Consider the following vector addition $\oplus$ and scalar multiplication $\otimes$, where $\lambda \in \mathbb{R}$:

$$\boldsymbol{x} \oplus \boldsymbol{y} = \boldsymbol{xy} \quad \text{and} \quad \lambda \otimes \boldsymbol{x} = \boldsymbol{x}^\lambda.$$

(a) Verify that $(\mathbb{R}^+, \oplus, \otimes)$ is a real vector space. State the zero vector and the negative vector of $\boldsymbol{x}$ for all $\boldsymbol{x} \in \mathbb{R}^+$.

(b) Interpret the property '$0 \otimes \boldsymbol{v} = \boldsymbol{0}$' in this real vector space.

Question 4 continued on next page...

Question 4. Consider the following 'law of cancellation' on a vector space $(\mathcal{V}, \oplus, \otimes)$ over $\mathbb{F}$ (where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$):

$$\text{for all } \boldsymbol{v} \in \mathcal{V},\ \ \lambda, \mu \in \mathbb{F},\ \text{ we have } \lambda \otimes \boldsymbol{v} = \mu \otimes \boldsymbol{v} \text{ implies } \lambda = \mu\,.$$

Does the above cancellation law hold for all vectors $\boldsymbol{v} \in \mathcal{V}$?
If it does not, can you impose additional condition(s) for the above cancellation to hold?

Question 5. Let $\mathcal{V}$ be a finite set with at least two elements. Can you define a vector addition $\oplus$ and scalar multiplication $\otimes$ so that $(\mathcal{V}, \oplus, \otimes)$ is a real vector space?

Question 6. Let $\mathcal{V} = \mathbb{C}$ and define $\boldsymbol{z} \oplus \boldsymbol{w} = |\boldsymbol{z}\boldsymbol{w}|$ and $\lambda \otimes \boldsymbol{z} = \lambda \boldsymbol{z}$ for all $\boldsymbol{z}, \boldsymbol{w}, \lambda \in \mathbb{C}$. Here, $\mathbb{C}$ is the set of complex numbers, and $|\boldsymbol{z}|$ is the modulus of the complex number $\boldsymbol{z}$.

(a) Write down each axiom in terms of the above operations $\oplus$ and $\otimes$. Identify which axiom holds or fails. If an axiom fails, give an example to illustrate it.

(b) Is $(\mathbb{C}, \oplus, \otimes)$ a complex vector space?

Topic: Subspaces

Question 7. Determine whether each of the following sets is a subspace of the real vector space $\mathbb{R}^3$ by checking whether it contains the zero vector, and it is closed under vector addition and scalar multiplication. Justify your answer.

(a) $\mathcal{W} = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 + 2a_2 - 3a_3 = 1\}$.

(b) $\mathcal{W} = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : 2a_1 - 7a_2 + a_3 \leq 0\}$.

(c) $\mathcal{W} = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : 5a_1^2 - 3a_2^2 + 6a_3^2 = 0\}$.

(d) $\mathcal{W} = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 = 3a_2, a_3 = -a_2\}$.

(e) $\mathcal{W} = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 + \sin(a_2) + a_3 = 0\}$.

Question 8. Consider the real vector space $\mathcal{F}(\mathbb{R}, \mathbb{R})$ of real-valued functions. Let d be a positive integer and let $\mathcal{W}$ be the set of all polynomials with degree *exactly* d. Is $\mathcal{W}$ a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$? Justify your answer.

Question 9. Consider the complex vector space $\mathcal{M}_n(\mathbb{C})$ of $n \times n$-matrices. For each matrix $\boldsymbol{A} \in \mathcal{M}_n(\mathbb{C})$, we define $\boldsymbol{A}^*$ as the *conjugate transpose* of $\boldsymbol{A}$, i.e., $\boldsymbol{A}^* = \overline{\boldsymbol{A}^T}$.
Is $\mathcal{W} = \{\boldsymbol{A} \in \mathcal{M}_n(\mathbb{C}) \mid \boldsymbol{A}^* = \boldsymbol{A}\}$ a subspace of $M_n(\mathbb{C})$?

Question 10. Consider the real vector space $\mathcal{M}_n(\mathbb{R})$ of $n \times n$ matrices. Which of the following subset $\mathcal{W}$ is a subspace of $\mathcal{M}_n(\mathbb{R})$? Justify your answer.

(a) $\mathcal{W} = \left\{\boldsymbol{A} \in \mathcal{M}_n(\mathbb{R}) : \boldsymbol{A}^T = -\boldsymbol{A}\right\}$.

(b) $\mathcal{W} = \{\boldsymbol{A} \in \mathcal{M}_n(\mathbb{R}) : \det(\boldsymbol{A}) \geq 0\}$.